March 13, 2025

Mr. Faisal D'Souza
NCO
2415 Eisenhower Avenue
Alexandria, VA 22314
Transmitted via email to ostp-ai-rfi@nitrd.gov.

Dear Mr. D'Souza:

Code.org submits the following in response to the Office of Science and Technology Policy (OSTP) and the NITRD NCO request for input on the Development of an Artificial Intelligence (AI) Action Plan ("Plan"), as printed in the *Federal Register* on February 6, 2025. This Action Plan, as directed by a Presidential Executive Order on January 23, 2025, will define the priority policy actions needed to sustain and enhance America's AI dominance, and to ensure that unnecessarily burdensome requirements do not hamper private sector AI innovation. Code.org and its partners believe that the safe, effective and responsible use of AI in schools comes from connecting the discussion of teaching with AI to teaching about AI. Any Action Plan must address the country's education and workforce development needs. Please consider our views on this crucial topic as you develop your Plan.

America demands economic opportunity.
Our working class – young and old, rural and urban – are being left behind by globalization, technological change, and a system that doesn't seem to reward hard work the way it used to. The American Dream feels broken.

Let's prepare people to navigate evolving work in the Age of AI.
AI is poised to change our economy even more than globalization. Before generative AI, automation was predicted to disrupt blue collar jobs. Now this trend is reaching both blue collar and white collar jobs. 60% of all workers believe AI will disrupt jobs. And the skills required to succeed in jobs are expected to change by 65% by 2030. In the past 10 years, jobs requiring high-end digital skills have grown while all other jobs have shrunk.

So does this mean we'll all be out of work? History shows the answer is no. In fact, AI might enable us to reach new heights. ***AI has the promise to be the next general purpose technology utilized across every job in every sector.*** In just one large-scale experiment, AI led to the discovery of 44% more materials, 39% more patents, and an overall R&D efficiency improvement by 15%.

We must prepare our current and future workforce for how work is evolving. AI will augment jobs as much as replace them.

Computer Science (CS) and AI education are the building blocks of our future workforce and a necessity for every student in our education system.
We are still early in AI's use, innovation, and advancement. Building trust in technology is just as important as building its usefulness. Trust comes from ensuring everyone has a foundational understanding of CS and AI. It shouldn't seem like magic to Americans. This knowledge will also ensure the future workforce can adapt and grow as the technology changes.

A lifetime $200K bonus in earnings for every American worker.
A groundbreaking new study from Brown University delivers data-backed proof that CS propels students to new heights. The economic benefits of high school CS education are profound. ***One high school CS course leads to an 8% increase in earnings at age 24 and a 3% higher likelihood of employment.***

If we want to restore the American Dream, why not start with CS and AI education to ensure every student can benefit from this opportunity? This can give a lifetime $200K bonus for every American worker[1].

Only 6.4% of US high school students take CS each year. Although over 60% of high schools offer high-quality CS courses, only 11 states require it to graduate. How is it that at the dawn of the Age of AI we still don't teach CS and AI to everyone? It's only common sense that if we were designing a school system from the ground up today, we'd start with CS and AI skills.

Every community benefits from this opportunity, not just the elite few.
Notably the 8% increase in earnings is found for every student, regardless of whether they go on to major in CS, whether they pursue a technical job, or even whether they attend college. This benefit isn't just for coastal tech hubs. It can spread to every community across America. In Pennsylvania alone, if every student took CS it could lead to an increase of $190M in ***annual*** earnings for each graduating class.

CS is even more important in the Age of AI.
Anyone can now use generative AI to write code. So some are asking "is coding dead?" In fact, learning about code along with AI unlocks a superpower only available to those that understand what the code is doing. The real questions are – do they understand what AI is creating? Can they apply it to create complex software to innovate and solve problems? That understanding is the heart of CS. That is what has driven the creation of the $2 trillion tech sector and established America as a global leader in AI.

[1] Assumes a $40,000 salary at age 24 increased by 8% and 2% lifetime wage growth yielding a lifetime earnings with CS of $2.7M and without CS of $2.5M.

Even as AI tools become more powerful to help with both of these questions, humans need to be able to evaluate and modify AI-generated code and understand the broader context and implications of programming. More than 70 tech and educational organizations leading the development of AI agree that learning to program still remains important.

AI promises to **democratize the ability to create software** by augmenting someone's technical knowledge with their domain-specific knowledge to solve problems in new and innovative ways. **But this is only possible with a foundational understanding of CS and AI.**

Given the cyber threat, we don't have a choice to wait.
Beyond the economic opportunity and the need to adapt in the Age of AI, threats from cybersecurity continue to loom. Expanding CS and AI education is a matter of *national security*. There are 500,000 open cyber positions in the US today, but this issue goes beyond these hiring challenges. The weakest link in our cyber defense is our own citizens and their inadequate technology education. Unless we take steps to better educate Americans in CS, our own safety is at risk.

Act now to keep America's Global Leadership.
Other nations are poised to overtake the United States as the worldwide leader in AI. In particular, China has been aggressively investing in AI research and development, and the National Security Commission on Artificial Intelligence warned that China is poised to overtake the U.S. within a decade. China has also invested heavily in education, putting CS and AI graduation requirements in place years ago. To counter these efforts and retain its competitive edge, the United States must invest in its future workforce by enhancing K-12 CS and AI education, ensuring the next generation is equipped with the skills necessary to drive AI innovation.

The solution lies in expanding CS and AI education.
Every student should take CS and AI. Here are three specific things the Trump administration can do to ensure our workforce can adapt and evolve in the Age of AI:

1. **Provide federal funding to states for K-12 schools to teach CS and AI.**
   Education is driven locally, but given the economic imperative, cyber security threats, and global competition for AI leadership, this is a shared federal and state issue. Consider that the US ***imports talent*** through H1B Visas, ⅔ of which are for software engineers and related occupations. This generates ~$500M/year.

   Every school should teach computer science and AI. During President Trump's first term, the Administration allocated $200M for CS from the Department of Education. This was a good start. This Administration should block grant +$500M each year back to states to expand CS and AI education and implement graduation requirements. ***This is just 1% of the US Department of Education's $50B budget!***

2. **Provide incentives to universities to prepare graduates for our workforce needs.**
   As tuition and student debt skyrocket, consider rewarding colleges that adapt their curriculum and teaching capacity to fit today's workplace needs. A CS degree, even from a lesser college, is worth much more than any other college degree[2]. Students with these degrees will repay their tuition loans faster and rarely default and should be incentivized.

3. **Expand retraining programs for adults.**
   Create programs to retrain adults in computing and AI skills to put the underemployed back to work. A recent IBM report estimated that 40% of the global workforce—equivalent to a staggering 1.4 billion individuals—will require reskilling over the next three years due to AI's impact. Teachers are no exception. The percentage of teachers who have attended an AI training session has risen from 13% in June 2023 to 43% in October 2024.

We don't need to throw more money at this problem.
The new Department of Government Efficiency is tasked with cutting federal waste and ensuring the government is spending its resources wisely. As a donor and grant-supported organization, Code.org leverages technology and has become the model of effective use of resources. We have more than 16M active students learning K-12 CS globally. We are the largest provider of high-quality K-12 CS courses in the US. And it costs us $6/student to actively engage them, and about $65/student to complete a high-quality high school CS course. That $65 can unlock an eight percent lifetime of earnings gains. We are a model that others can look to on how to effectively scale and reach millions of students.

Everybody agrees we should do this.
Expanding K-12 CS and AI education has the strong support of Governors, members of Congress, and mayors from both sides of the aisle. Surveys show that **90%** of American parents want their children to study computer science in school. A coalition of 800 CEOs signed an open letter urging governors to expand computer science education. This issue has the support of not just every major *tech* CEO, but also the CEOs of America's top airlines, hotel chains, banks, manufacturers, entertainment companies, you name it.

Americans of all backgrounds and political persuasions are finding common ground on this issue. The American Dream is broken. Giving every American worker a chance to earn a lifetime $200K bonus through CS education is one of the best ways to fix it.

Hadi Partovi	**Cameron Wilson**
CEO, Code.org	**President, Code.org**

[2] Comparison of the lifetime value of college degrees: the Hamilton Project.

About Code.org
Code.org® is an education innovation nonprofit dedicated to the vision that every student in every school has the opportunity to learn computer science and artificial intelligence as part of their core K-12 education.

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